

Why We Should Study Cancer Like We Study Ecosystems

By Rachel Nuwer

When pine beetles invaded North American forests, they **threw** a delicate ecosystem **out of balance**ⁱ. Cancer cells, too, behave like **damaging**ⁱⁱ **invasive** species.

- i. Throw sb/sth out of balance means to make sth unsteady, 打破平衡;
- ii. Damaging means destructive, 有破坏力的;

Sometimes, thinking about an old problem from a **refreshing** new **angle**ⁱⁱⁱ is just the thing needed to find that eureka moment.

- iii. 我们做一个词形辨析:
 angle: aspect, side, facet, 角, 角度;
 angel: a spiritual being, 天使;
 anger: noun form of angry.

Cancer, one of the most **notorious**^{iv} medical **maladies**(malady = disease, illness), has been studied **intensely**^v in the current era of modern medicine. But a growing number of researchers think that bringing a fresh, out-of-the-box **approach**(=way) to understanding the disease may lead to some **novel** insights and, perhaps, solutions. And the subject that they're hoping can serve as a window into the study of cancer may surprise you: ecology.

- iv Notorious means widely known especially for some bad quality;
- v To study intensely means to research profoundly, deeply 深研;

On face value(=From outward appearance, **On the surface** 从表象上看), **oncology**(肿瘤学) and ecology seem vastly different. For example, one is **localized**^{vi} to specific cells in the body, while the other by definition **spans**^{vii} the entire globe. But rather than **labeling** cancer as a group of **mutated** cells, as the thinking goes, we should see cancer as a **disruption**^{viii} in the balance of a complex microenvironment in the human body. Like a **damaging invasive** beetle **eating its way through**^{ix} forests in Colorado, a novel disease **breaking out**^x in populations of wild birds, or loggers **mowing down**^{xi} parts of the Amazon rainforest, cancer **throws a monkey wrench into**^{xii} an **otherwise**^{xiii} **placid**, balanced system.

- vi. Localise means to situate, locate, position.
- vii. Span means to spread, extend wide across 跨越;
- viii. Disruption means interruption, disturbance, disorder, chaos;
- ix. 和 eat one's way through(一路吃过去)结构相当的, 还有很多其他表达, 如
 Dance one's way to, e.g. She danced her way to a renown artist. 她舞成了一位知名艺术家。
 He speaks his way to influence. 他凭一张嘴打天下。
 Behrman studies his way to expertise. 伯曼学成了一位专家。
 He runs his way to victory. 他一“跑”成名。
- x. 我们将具体行为动作 A novel disease breaks out 写成抽象化概念: the **outbreak** of a novel disease
- xi. Mow down means to kill or destroy in great numbers.



- xii. This is a monkey wrench: 英语叫它“猴钳”，咱们叫“老虎钳”。
Throw a monkey wrench in/into means to disrupt, to cause problem to a plan, activity or project. 打破平静，使混乱；
- xiii Here, otherwise means under other circumstances. The sentence can be paraphrased like this: Under circumstances where no mutated cells intrude, our body environment would be in a perfectly placid balance.(Under other circumstances 中的 other, 在该句中表示在没有肿瘤侵犯的情形下)

This way of thinking makes cancer seem even more complex than it already is, but it could provide **insights**^{xiv} that **ultimately**^{xv} make cancer more **treatable**, **propose** researchers from the Moffet Cancer Center in a paper published in the journal *Interface Focus*.

- xiv. Insight here means new ways of understanding or perception;
xv. Ultimately has many synonyms, like eventually, finally, in the end, at last;

“Einstein is known to have said that everything should be made as simple as possible, but not simpler,” they write. “It **turns out**^{xvi} that **complexity** has its place and, **as convenient as**^{xvii} it would be for cancer biologists to study tumor cells in **isolation**, that makes as much sense as trying to understand frogs without considering that they **tend to** live near **swamps**(沼泽) and **feast on** insects.”

- xvi Turn out means to be found to be, as after experience or trial.
xvii As convenient as here means though.

We tend to think of cancer only **in terms of** mutated cells, the authors continue. But adopting this narrow approach is like trying to understand why a frog has a **sticky** tongue without taking into account that frogs use their tongues to catch insects. Cancer cells, **likewise**, need context. A **voracious** cancer cell, for example, may situate itself next to a blood vessel not **by chance**, but **by choice** so it can obtain more **nutrients** and oxygen to support its unlimited **division**(分化).

- xviii. In terms of means regarding, concerning, on the basis of 从某角度看；
xix. A voracious cancer cell may situate itself next to a blood vessel not by chance but by choice... The sentence can be paraphrased into: A **greedy**(贪婪的) cancer cell will not position itself **accidentally** but **deliberately** near a blood vessel.

Cancer cells must **compete** within the body **for** nutrients and other resources, just like animals living in an environment must compete with one another in order to survive. This means that cancer, like any organism, must adapt to its environment in order to **thrive**(develop, flourish 生长). The researchers explain:

*It is now beginning to be widely accepted that cancer is not just a genetic disease but the one in which **evolution** plays a crucial role. This means that tumor cells **evolve**, adapt to and change the environment in which they live. The ones that fail to do so will ultimately become **extinct**^{xx}. The ones that do will have a chance to **invade** and metastasize. The **capacity**^{xxi} of a tumor cell to adapt to a new environment will thus be **determined** by environment and the cellular species from the original site, to which it has already **painstakingly**^{xxii} (费力地) adapted.*

xx. Extinct is an adj.; it means no longer active or existing. Its verb form is extinguish.

我们做一个**词形辨析**：

Extinguish. E.g. We have light here. Extinguish the fire. 有光了。把火熄灭了。

Distinguish. E.g. Can you distinguish tumor cells from non-cancerous cells? 区别，区分。

xxi. Capacity means ability, capability 能力；

xxii. 注意拼写，不能写成 painttaking。

So how can all of this theory be applied in real life? The environmental approach to understanding cancer is so complex that it rules out^{xxiii} normal experiments; they could easily go awry with^{xxiv} so many different components to consider. Instead, the researchers suggest^{xxv} turning to mathematics and computation for understanding the greater environmental context that leads to cancer. Ecologists use one such mathematical approach, game theory, as a way to study evolutionary biology and the way animals interact:

The force of natural selection keeps ecosystem denizens^{xxvi} focused on optimizing^{xxvii} the bottom line^{xxviii}: long-term reproduction. In the games studied by evolutionary game theoreticians, individuals compete for available resources using a variety of strategies. These features and behaviors, known as the phenotypic strategy, determine the winners and losers of evolution.

xxiii. Rule out simply means to remove from consideration, exclude, 排除某物的可能性。“The environmental approach to understanding cancer is so complex that it rules out normal experiments” means that the environmental way to prove if this theory is applicable or not cannot be carried out in experiments. 也就是说，从生态角度推演出肿瘤细胞的行为策略是无法通过实验来证明的。

xxiv. Go awry with means to go away from the correct course, 走歪，偏道；

xxv. 老师这里汇总 suggest 的用法：

- 1) 表示暗示，说明，其用法和 indicate, imply 类似。E.g. This suggests that he is an insider. 这说明他是个内线；
- 2) 表示建议，提议。E.g. I suggest that we (should) go to Sun Yat-sun Mausoleum tomorrow. Or I suggest Sun Yat-sen Mausoleum as our next touring site. Or I suggest going to Sun Yat-sen Mausoleum during Labor's Day.

xxvi. Denizen refers to a resident or an inhabitant. This can be a human, or an animal.

xxvii. Optimise means to make as perfect or effective as possible, to make the full advantage of 最佳化，最优化，充分利用；

xxviii. The bottom line refers to the most important or essential point, 表示最重要的事情。同学们注意，坚守底线，是无法用 the bottom line 来表达的。大家平时要培养查阅英英释义词典的习惯。这里提供几种“坚守底线”的表达法：

Stick to my principles, 坚持我的原则；Stick to or uphold or maintain the standards, 维持标准；Follow/Obey the rules. 遵循原则。

Behavioral strategies may change depending upon both an animal's nature and the situation's context. Here's a hypothetical(假设性的) example, based upon game theory thinking: If two hyenas are digging into a large, tasty wild beast carcass, they'll

happily share that resource. But if two lions find that same carcass, they will fight for **exclusive** rights to eating it, meaning one lion **emerges victorious** and takes all the **meaty spoils**(=prey 猎物), while the other gets no food or even gets injured. Finally, if a lion meets a hyena at the carcass, the hyena will bolt, **surrendering**(=giving up) its goods **to** the stronger lion. In other words, game theory players can react one of the three ways depending upon who they are and what's going on: they can share, fight or forfeit(=give up, surrender).

xxix. Tasty means having a pleasing flavor, 美味的, 和 juicy 词以接近, 美味多汁的;

xxx. Wild beast, 野兽; 兽性: bestiality;

xxxi. Victorious means successful; it is the adj. of victory. 胜利;

xxxii. Meaty means heavily fleshed, 肥硕的;

Similar games may be played with tumor cells. "A good example would be a tumor with cells that move away when **confronted with scarce**(short in supply 稀少的) resources and cells that stay to use them," the authors write. To make things even more complicated, however, tumor cells are known to change their behavior as they **proliferate**(to increase at a rapid speed) and metastasize(to transmit, spread) throughout the body, meaning they could **switch** from a hyena to a lion.

One **crucial**(=significant, important) thing that game theory at an ecosystem level shows us, they continue, is that **indiscriminately**^{xxxiii} focusing on killing as many tumor cells as possible might not provide the best **outcome**(=result) for the patient. According to game theory models, the **eventual long-term**^{xxxiv} result of the game depends upon specific interactions between the players, not on the number of players involved. Lions will continue to fight one another for food, **regardless of** whether two lions or 2,000 lions meet. "A treatment based **exclusively**^{xxxv} on indiscriminately **removing** most (but not all) cancer cells may only have a **temporary**^{xxxvi} effect; as in most cases, the original number of tumor cells will eventually be **restored** and **exceeded**^{xxxvii}," the authors write.

xxxiii. Discrimination 表示歧视, 区别对待; 那么, indiscriminately 就表示不加以区分地, 不分青红皂白地;

xxxiv. 在 longtime 和 long-term 的词义区分上, 一直都存在差异。老师这里尝试讲一下它们的区别: Longtime 表示长久的, 这种时间的长久不存在任何影射, 例如 a longtime friend, 老朋友一个。但 long-term 表示某种影响持续的长久, it means **lasting or being in effect for** quite a long period of time.

我们既有 a longtime friendship, 也有 a long-term friendship. 但前者纯粹修饰友谊时间的长短, 而后者侧重友谊的影响深远。

xxxv. Exclude means to prevent or bar from participation, consideration or inclusion, 表示把某物排除在外; exclusively 表示排他地, 专门地, 在原文里基本等于 solely, simply, only. "A treatment that is based only/solely/simply on killing cancer cells..."

xxxvi. 我们做一个词形辨析:

Temporary means lasting for a very short time;

Contemporary means modern, current 当代的, 现代的

xxxvii. Exceed has many many synonyms, like **surpass**, **outperform**, **outstrip**,

transcend.

Instead, game theory indicates that a more effective **alternative(=choice)** would be based on trying to change the ways that cells interact with one another and with their environment. This may **affect(strongly influence)** the cells' behavior, strength and reproductive success, the authors explain, which could drive a tumor's evolution towards less **aggressive** cell types, or to a more stable **coexistence** with non-cancerous cells.

"The ecosystem view is, **ultimately**, a **holistic(=complete, comprehensive 全面的)** one that sees cancer progression as a process that emerges from the interactions between **multiple cellular** species and interactions with the tumor microenvironment," the authors write. "An ecosystem **perspective** presents us with **intriguing implications^{xxxviii}**," they say, **along with(together with) a host of(=a great number of)** questions about how far the **analogy** between ecosystem and cancer can be taken.

xxxviii. intriguing means engaging, attention-absorbing; implication means significant possibilities 潜在的重要的可能性;

For example, if cancer cells spread like an invasive species through an ecosystem, what **evolutionary** gain is achieved when the closed off ecosystem (a body) is **irreparably^{xxxix}** damaged (through a person's death) such that the **pestilence(pest, plague 瘟疫)** also dies? Unlike a **virus**, which may kill its host but spread to other hosts in the process, cancer cells themselves, for the most part, have no means of spreading from individual to individual. And are cancer cells taking their **cues(=signals, hints)** from processes driven by competition or from cooperation? Thinking more **proactively**, can non-cancerous cells be **triggered(=motivate, activate, set off 激活, 启动)** so that they behave like lions and usurp(=seize and make use of 篡夺) cancerous cells' resources until the cancer is **manageable^{xxxxi}**?

xxxix. Repair 表示修补; repairable means capable of being repaired, 可修补的; irreparably means irreversibly, 不可逆转的;

xxxx. Pro- means forward, toward the front, 此前缀表示向前, 往前; Proactive means acting in anticipation of future problems, needs or changes. 有远见性质地活动的, 以解决未来问题、需求等为目标的活动。有的近义词有 foreseeing, farsighted, 有前瞻性的, 有原件的, forethoughtful, 提前做思考的。

xxxxi. Here, manageable means capable of being handled, dealt with or treated.

While ecology and mathematics likely will not **defeat** cancer on their own, viewing the disease from this perspective could allow doctors to better **predict** where in the body tumor cells have the best and worst chances of survival, and how to most effectively prevent them from **proliferating**.

"The heart of the matter is that an ecological view of tumors does not **invalidate^{xxxxii}** but **complements^{xxxxiii}** and builds upon **decades** of cancer research and **undoubtedly** this will lead to a better understanding of the biology of cancer and to new and improved **therapies**," the researchers conclude. "We need to **properly(=correctly, accurately 正确地)** understand the trees (e.g. every leaf, twig and branch) before we can understand the forest but we **cannot^{xxxxiv} afford to ignore** the forest because the trees are so

interesting on their own.”

xxxii. Valid means effective, logical, correct, justified. So invalidate means to make sth invalid, or incorrect, ineffective. 使无效;

xxxiii. 我们做一个词形辨析:

Complement means to make sth complete, 表示使某物完整, 补充;

Compliment means to praise, to appreciate or admire 褒奖; It can also be a noun.

Comprehensive means all-embracing, complete, 全面的, 完整的, 综合的;

Comprehend means to understand;

xxxiv. 这里, 给同学们再次强调 cannot 的正确书写。我们说“不能”或者“无法”, 它的正确形式是 cannot, 而不是 can not. 例如, 他们不会说汉语。“不会”表示没有能力, 表达是: They cannot speak Chinese. 而不是 They can not speak Chinese. **DO NOT** separate “can” from “not”. 或者说, can“能”的反义词是 cannot, **而不是 can not.**